

WireShark

Cybersecurity Internship – Phase 1, Task 6

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Internship Program: Cybersecurity Internship – UJAR TECH

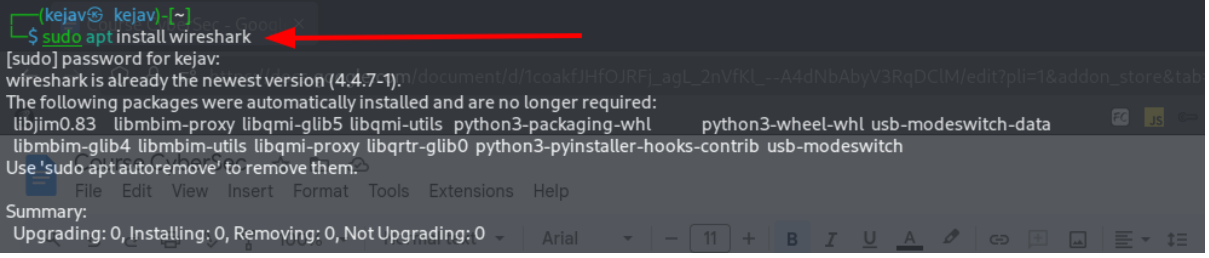
What is Wireshark?

Security professionals and system administrators widely use Wireshark, an open-source network packet or protocol analyzer. They consider it an essential tool. Analysts use it to examine the structure of different network protocols and demonstrate applications. Users can operate Wireshark on various platforms such as Windows, Unix, and Linux, and it employs the GTK+ widget toolkit or PCAP for packet capturing. IT also has terminal-based free software versions like Tshark. Wireshark shares many characteristics with tcpdump only the difference is that it supports a graphical user interface (GUI) and has information filtering features.

Installation of Wireshark

To download Wireshark, open a terminal and type the following command to install Wireshark:

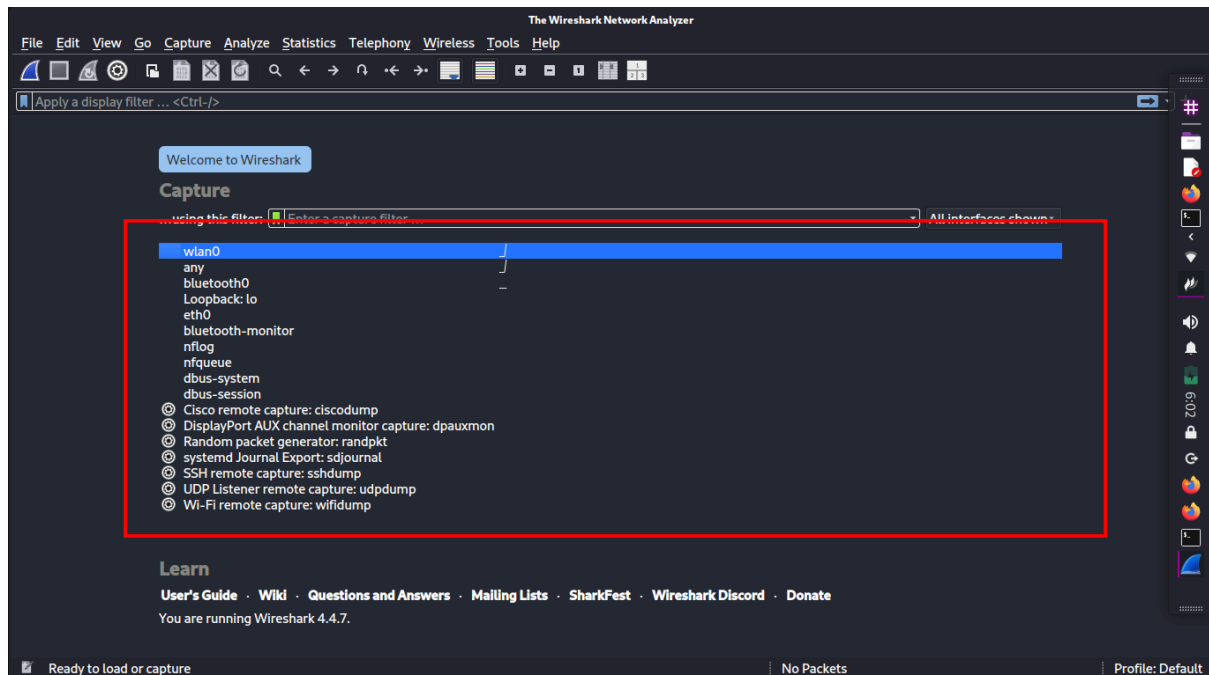
- **apt install wireshark**



```
(kejav@kejav:~)$ sudo apt install wireshark
[sudo] password for kejav:
wireshark is already the newest version (4.4.7-1).
The following packages were automatically installed and are no longer required:
  libjim0.83 libmbim-proxy libqmi-glib5 libqmi-utils python3-packaging-whl python3-wheel-whl usb-modeswitch-data
  libmbim-glib4 libmbim-utils libqmi-proxy libqtr-glib0 python3-pyinstaller-hooks-contrib usb-modeswitch
Use 'sudo apt autoremove' to remove them.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

How to use Wireshark?

Whenever you open Wireshark, you will be prompted with the following screen.



Here you can see different network interfaces on your device. In the above image, we can see that many devices communicate a lot of traffic through the Wi-Fi interface. In most cases, you will only be able to see traffic going in and out of your device; however, some

wireless network cards can enter monitor mode so that you will be able to see traffic from other wireless devices on the network.

Packet Capturing

The following methods can be used to start capturing packets. You can double-click on the interface in the welcome screen of Wireshark. This will start Wireshark capturing on interface wlan0. Once you capture some packets, you can view the packets displayed in the packet list pane by simply clicking on a packet in the packet list pane, which will bring up the selected packet in the tree view and byte view panes. As soon as you capture some traffic, you need to apply a filter to make it easily understandable.

Wireshark has two filtering languages

- **Capture filters**
- **Display filters**

You use capture filters for filtering when capturing packets, and you use display filters for filtering which packets are displayed. Wireshark provides a display filter language that enables you to have precise control over which packets are displayed.

capture filters

[Capture filter\[23\]](#) limit the captured packets by the chosen filter. If the packets don't match the filter, Wireshark won't save them. Examples of capture filters include:

host IP-address: This filter limits the captured traffic to and from the IP address

net 192.168.0.0/24: This filter captures all traffic on the subnet

dst host IP-address: Capture packets sent to the specified host

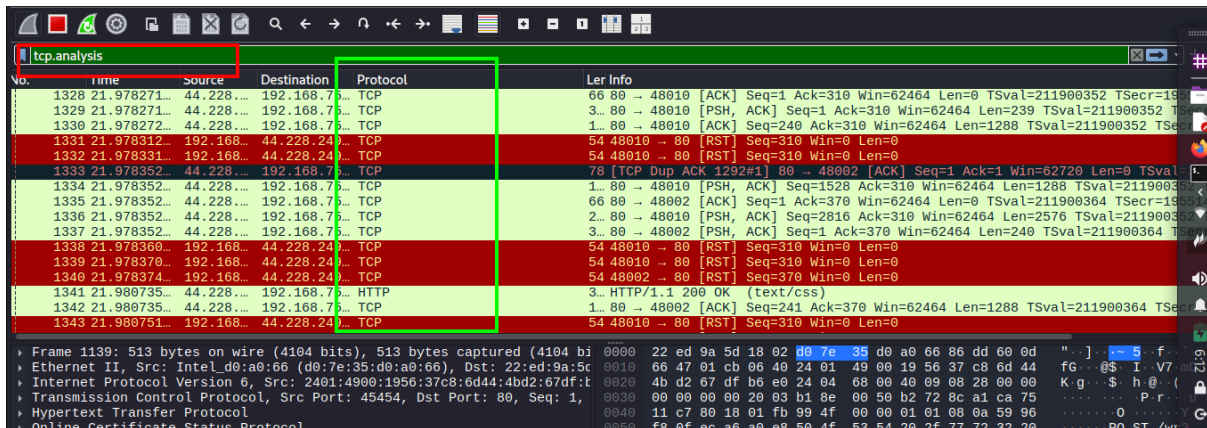
port 53: Capture traffic on port 53 only

port not 53 and not arp: Capture all traffic except DNS and ARP traffic

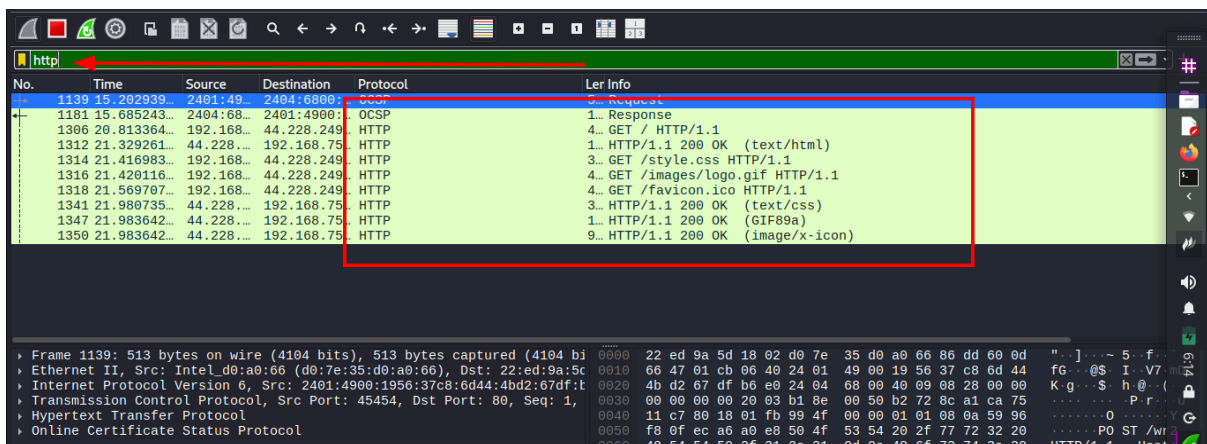
Display filter

Wireshark's display filters are located right above the column display section. To only display packets containing a particular protocol, type the protocol into Wireshark's display filter toolbar. Wireshark offers a list of suggestions based on the text that you typed.

For example, to only **display TCP packets**, type tcp into Wireshark's display filter toolbar.



Similarly, to only display packets containing a particular field, type the field into Wireshark's display filter toolbar. For example, to only display HTTP packets, type **http** into Wireshark's display filter toolbar and it will accept the expression and works as intended



As you saw above the expression works but not intended. As we have noticed these packet captures have different colours. So, what are these colours intended for...? Don't get confused with a different type of colour packets. These colours are intended for

- Gray – TCP packets
- Black with red letters – TCP Packets with errors
- Green – HTTP Packets
- Light Blue – UDP Packets
- Pale Blue – ARP Packets
- Lavender – ICMP Packets
- Black with green letters – ICMP Packets with errors

Note: – Colourings can be changed under View -> Colouring Rules

Building Display Filter expressions:

We can build display filters that compare values using a different type of comparison operator.

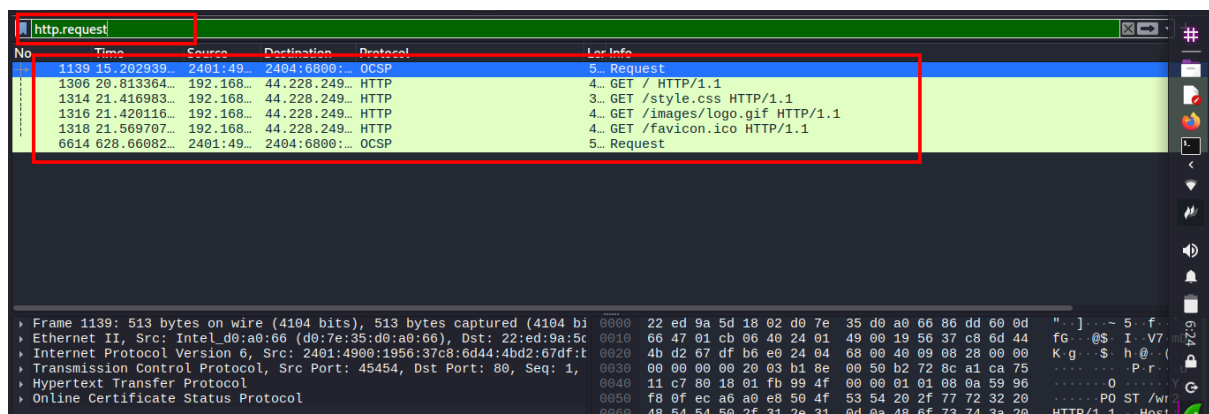
For example, to display only packets to or from the IP address 10.96.200.253, use **ip.addr==10.96.200.253**. Wireshark display filter uses Boolean expressions, so we can

specify values and chain them together. A complete list of available comparison operators is shown below.

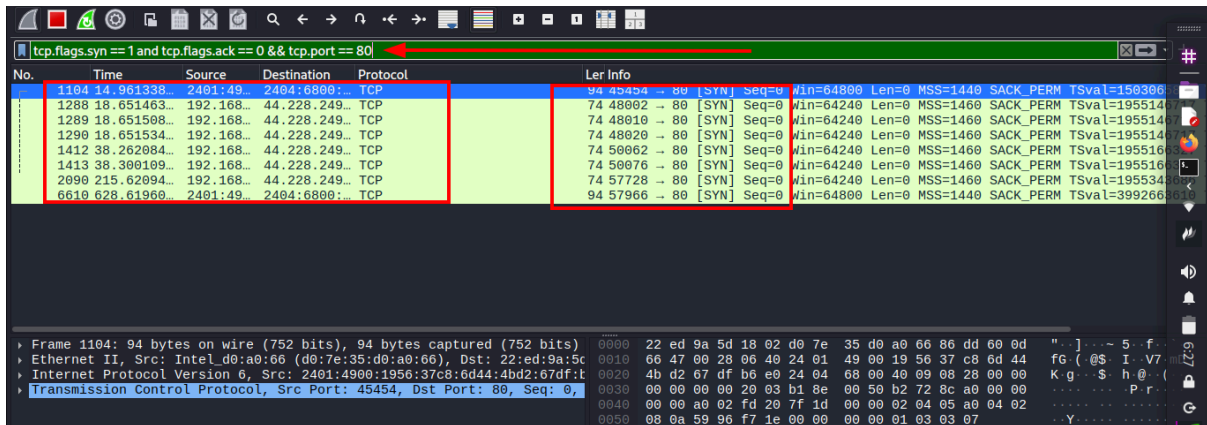
Description	operator	Example
Equal or (eq)	==	ip.src==192.168.0.134
Not equal or (ne)	!=	ip.src!=192.168.0.134
Greater than or (gt)	>	frame.len > 20
Less than or (lt)	<	frame.len < 150
Greater than or equal to or (ge)	>=	frame.len >= 0x50
Less than or equal to or (le)	<=	frame.len <= 0x30
Bitwise_and	&	tcp.flags & 0x03
Logical AND or and	&&	ip.src==192.168.0.134 and tcp.flags.fin
Logical or		ip.src==192.168.0.134 or ip.src==192.168.1.1

Some Useful Filters

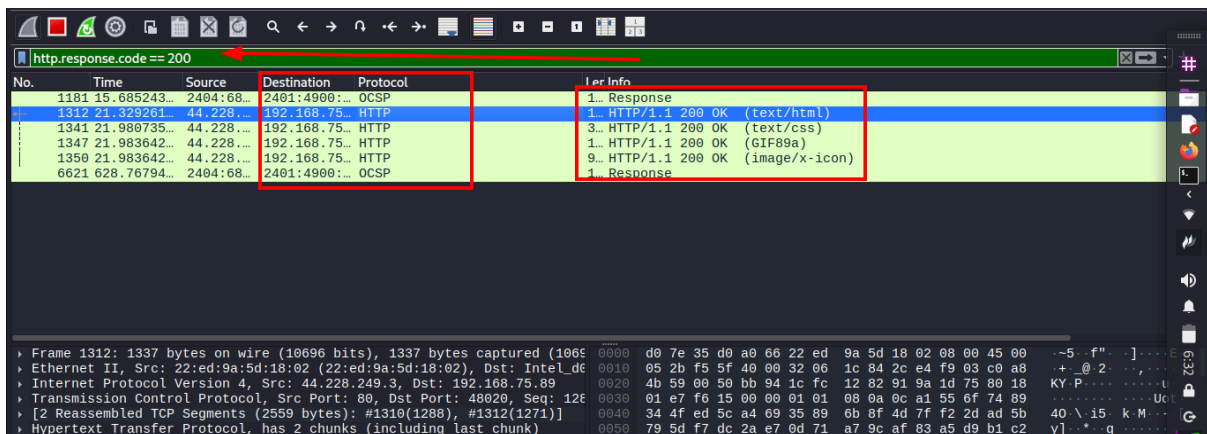
Here are some filter expressions that you can use to quickly review web traffic. Let's understand this with some sort of method, like how we are going to filter some infectious traffic. Open the packet capture and apply the following filter: **"http.request"**. This filter will show all HTTP post requests. Also, you can find the total no. of packets at the bottom of the Wireshark screen that are 7 of these packets.



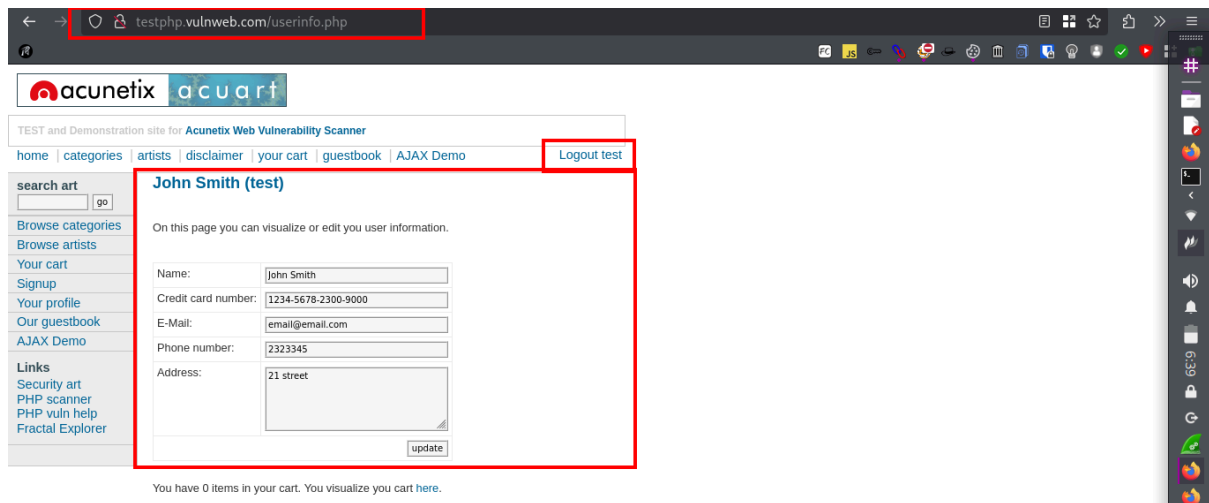
To reveal all the TCP syn packets, we can use the following expression as a way to quickly review web traffic for port 80. Also, you can find the total no. of packets at the bottom of the Wireshark screen, which are 8 of these packets.



The value **http.response** reveals URLs for HTTP responses, and **HTTP status code 200** means success. The client has requested documents from the server. The server has replied to the client and given the client the documents. Filtering with this display filter can outline the flow of events for the web traffic.



When we log in to a website that uses HTTP, the data is sent in plain text. For example, I tested this on a demo website where I logged in and captured the traffic using Wireshark. I noticed that the login request was being transmitted in plain text. This means that if we are using public Wi-Fi, such data can easily be intercepted, so we must be very careful in such situations.



Captured the traffic using Wireshark

